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Oil Volatility Risk

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1 Big picture

- **Question.** Does oil-price volatility contain a distinct macro-financial risk factor, separate from equity-market volatility and general macroeconomic uncertainty?
- **Main empirical object.** The paper studies option-implied oil variance and asks how it predicts future macroeconomic growth, oil-sector quantities, and asset returns.
- **Core empirical finding.** Implied oil variance is a strong negative predictor of future GDP growth, consumption growth, investment growth, and employment growth. It also predicts higher oil inventories and lower oil consumption and oil production.
- **Not just generic uncertainty.** Oil volatility remains economically important when controlling for implied equity variance and a macroeconomic variance measure. Equity and macro variances do not fully substitute for oil variance.
- **Asset-price implication.** Industries and portfolios differ in exposure to oil variance. Aggregate market returns and many non-oil industries have negative oil-variance exposures, while the oil sector has positive oil-variance exposure.
- **Mechanism.** The authors build a production economy with an oil sector and inventories. Oil supply uncertainty induces precautionary oil storage. This reduces effective oil available for production and lowers macroeconomic output, consumption, investment, and employment.
- **Model role.** The quantitative model shows that oil uncertainty shocks can produce the empirical sign pattern: higher oil inventories, lower oil consumption, lower macro activity, lower aggregate equity returns, and positive returns for oil-sector equity.
- **Economic takeaway.** Oil volatility is not merely a financial-market symptom. It works like a real-sector uncertainty shock to a crucial production input. The macro effects are strongest when oil is difficult to substitute away from and when precautionary inventory demand is important.

2 Data and measurement

2.1 Sample and frequency

- The benchmark sample is quarterly from 1990Q1 to 2019Q4.
- The paper combines oil-market data, macroeconomic aggregates, uncertainty measures, and asset returns.
- Several exercises also use monthly data for oil futures basis and asset-market outcomes.

Interpretation: The paper is designed to distinguish oil-specific uncertainty from aggregate financial uncertainty. Quarterly data line up with macro variables, while monthly data provide finer evidence on oil-market pricing and futures-basis dynamics.

2.2 Main uncertainty measures

- **Oil variance:** option-implied variance from oil options, scaled as an uncertainty measure.
- **Equity variance:** option-implied equity-market variance.
- **Macro variance:** a macroeconomic variance measure based on productivity or macroeconomic uncertainty.

Interpretation: The key empirical test is whether oil variance predicts outcomes after controlling for equity and macro variance. If oil variance only reflected generic market stress, its coefficient should disappear in multiple-volatility regressions.

2.3 Macro variables and oil-sector variables

- Macro variables include GDP growth, consumption growth, investment growth, employment growth, and TFP growth.
- Oil-sector variables include oil inventory growth, oil consumption growth, and oil production growth.
- Asset-price outcomes include oil returns, aggregate equity returns, industry portfolio returns, dividends, and forecasts or uncertainty indexes.

Interpretation: The variables are chosen to separate two channels: a macro production channel and an oil-storage channel. The model predicts both lower macro activity and higher precautionary inventories after oil uncertainty rises.

2.4 Oil futures basis

- The oil basis is based on the difference between spot and futures prices.
- The paper analyzes two-, six-, and twelve-month maturities.
- A negative relation between oil inventories and oil basis is a standard prediction of storage theory.

Interpretation: The futures basis is useful because it is a market-based proxy for the shadow value of inventories and expected scarcity in the oil market.

2.5 Industry portfolios

- The paper estimates exposures of the aggregate market and Fama-French industry portfolios to oil variance and oil price risk.
- The oil industry tends to have positive oil-variance exposure, while many other sectors have negative exposure.

Interpretation: Cross-sectional return exposures help validate the proposed channel. If oil uncertainty depresses non-oil production but raises the relative value of oil-sector claims, returns should show opposite signs across sectors.

3 Empirical specifications and model equations

3.1 Predictive regressions

The baseline empirical specification can be written as

$$\sum_{j=1}^h y_{t+j} = a_h + \beta'_h V_t + \gamma'_h z_t + \varepsilon_{t+h}, \quad (1)$$

where y is a macroeconomic, oil-sector, or asset-price outcome; V_t is the vector of volatility variables; and z_t contains controls such as current and lagged oil returns, oil supply growth, and TFP growth.

Interpretation: The coefficient on oil variance asks whether oil-specific uncertainty predicts the outcome at horizon h after controlling for current conditions and other uncertainty measures.

3.2 Single-volatility versus multiple-volatility evidence

- In single-volatility regressions, each uncertainty measure is entered separately.
- In multiple-volatility regressions, oil variance, equity variance, and macro variance enter together.

Interpretation: The multiple-volatility regressions are the cleanest evidence that oil variance is distinct. If oil variance remains significant when other volatility variables enter, it captures an oil-specific propagation channel rather than generic uncertainty.

3.3 Impulse responses

The paper estimates VAR-based impulse responses to positive oil-variance shocks. The ordering is varied so that oil variance is ordered both early and late among the variables.

Interpretation: The two orderings ask whether results depend on a strong contemporaneous causal ordering assumption. Similar negative responses across orderings support the robustness of the result.

3.4 Production economy

The model features a final-goods producer that uses labor, general capital, and oil. A stylized version of the production function is

$$Y_t = (A_t N_t)^{1-\alpha} \left[(1 - \bar{l}) K_t^{1-1/o} + \bar{l} J_t^{1-1/o} \right]^{\frac{\alpha}{1-1/o}}, \quad (2)$$

where A_t is productivity, N_t is labor, K_t is capital, J_t is oil input, and o governs substitution between oil and capital.

Interpretation: The elasticity parameter is central. When oil is hard to substitute away from, uncertainty about effective oil supply has a larger macroeconomic impact.

3.5 Precautionary inventory channel

Oil inventories smooth oil-supply fluctuations. When oil supply uncertainty rises, agents hold more inventories to insure against future stock-outs. This reduces oil available for immediate production.

Interpretation: This is the model's distinctive channel. Higher uncertainty raises precautionary inventories, but the inventory buildup makes effective oil supply tighter today, lowering output and investment.

3.6 Simulation regressions

The model is simulated and the same predictive regressions are run on simulated data:

$$y_{t+1} = a + \beta' V_{t+1} + \gamma' z_{t+1} + e_{t+1}. \quad (3)$$

Interpretation: The goal is not only to fit moments but also to reproduce the regression sign pattern: oil variance should predict negative macro growth and oil consumption, positive inventories, and opposite asset-return exposures across oil and non-oil sectors.

4 Identification and empirical design

4.1 What identifies the empirical relation

- Identification is based on time-series variation in oil option-implied variance and its relation to future macroeconomic and sectoral outcomes.
- Regressions control for current and lagged oil returns, oil supply growth, TFP growth, equity variance, and macro variance.
- The paper also considers rolling-window estimates, VAR impulse responses, forecasts, uncertainty indexes, and cross-sectional asset exposures.

Interpretation: This is not a randomized experiment. Credibility comes from consistency across many outcomes and from the fact that oil variance survives comparisons to other volatility measures.

4.2 Why the Great Recession robustness matters

- The Great Recession was a period of extreme financial and macro stress.
- If oil variance only captured generic crisis uncertainty, removing 2007Q4–2009Q2 should eliminate the predictive relationship.
- The paper shows the oil-variance results are similar when the Great Recession is excluded.

Interpretation: This check supports an oil-specific mechanism rather than a purely financial-crisis explanation.

4.3 Why the model evidence matters

- The empirical patterns alone show predictability but do not explain the propagation channel.
- The quantitative model connects oil uncertainty to inventories, effective oil supply, output, consumption, investment, and asset prices.
- Simulated regressions then test whether the model can reproduce the empirical sign pattern.

Interpretation: The model is a mechanism test: it asks whether a plausible production economy with oil inventories can rationalize the data.

4.4 Main limitations

- Option-implied oil variance may reflect both physical uncertainty and risk premia.
- Time-series regressions may still be affected by omitted global shocks.
- The model is stylized and abstracts from many institutional details of oil markets, financial intermediation, and geopolitical events.
- Quarterly data smooth high-frequency oil-market news and may blur contemporaneous causal timing.

Interpretation: The paper's contribution is not a single clean quasi-experiment. It is a broad empirical and model-based case that oil uncertainty is a distinctive macro-financial risk factor.

5 Tables and figures

5.1 Figure 1: Time series of excess returns and uncertainty measures; Table 1: Summary statistics

Read:

- Figure 1 shows that oil returns are volatile and that implied oil variance spikes around major energy and macro events.
- Oil variance is much larger in scale than equity variance and is not simply a mirror image of equity-market volatility.
- Table 1 shows that oil variance is negatively correlated with GDP growth, consumption growth, investment growth, and employment growth.
- Oil variance is positively related to oil inventories and negatively related to oil consumption and oil production, consistent with a precautionary storage channel.

Economic message: Oil uncertainty co-moves with weak macro activity and oil-market adjustment. The summary statistics motivate the main regressions by showing that oil volatility is not just a financial variable but is linked to real activity and oil-sector quantities.

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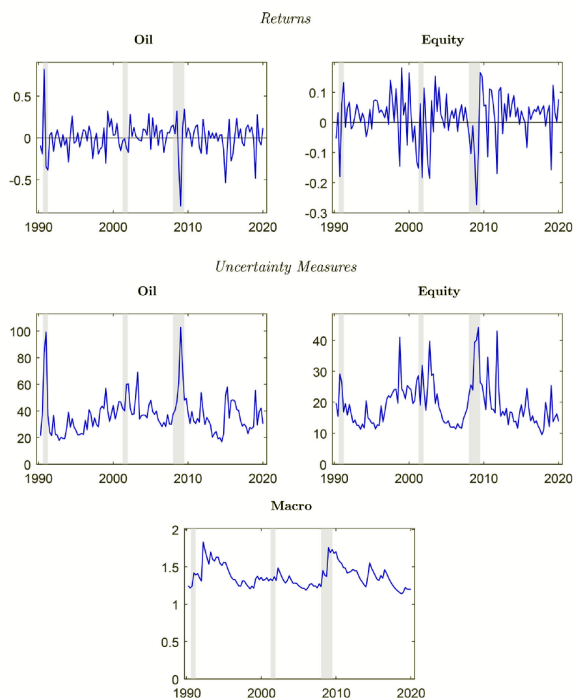


Figure 1: Time series of excess returns and uncertainty measures. The figure shows the time series of excess returns in oil and equity markets (top panel) and macroeconomic uncertainty (bottom panel). For ease of exposition, volatility series are expressed in annualized standard deviation terms. Vertical shaded regions indicate recessions as defined by the NBER. Data are quarterly from 1990Q1 to 2019Q4.

(a) Figure 1: returns and uncertainty measures

Table 1

Summary statistics.

	Mean	Std Dev	Aut(1)	Quarterly Correlations with			Annual Correlations with		
				Oil Var	Eq Var	Macro Var	Oil Var	Eq Var	Macro Var
GDP Growth	1.50	1.15	0.38	-0.57	-0.41	-0.21	-0.64	-0.54	-0.51
Consumption Growth	1.35	0.84	0.35	-0.36	-0.34	-0.19	-0.51	-0.47	-0.47
Investment Growth	3.23	3.72	0.66	-0.51	-0.36	-0.23	-0.46	-0.40	-0.39
Employment Growth	0.17	0.87	0.86	-0.59	-0.54	-0.33	-0.45	-0.50	-0.47
TFP Growth	0.79	1.41	-0.02	0.04	0.11	0.02	-0.11	0.05	-0.17
Oil Inventory Growth	0.47	2.29	0.20	0.08	-0.01	0.06	0.24	0.02	0.20
Oil Consumption Growth	0.09	2.39	-0.19	-0.15	-0.07	0.04	-0.79	-0.57	-0.32
Oil Production Growth	0.98	2.36	0.02	-0.04	-0.04	-0.07	-0.27	-0.03	-0.12
Excess oil return	0.69	38.36	-0.07	-0.27	-0.16	-0.07	-0.32	-0.30	-0.16
Excess equity return	6.49	16.22	0.01	-0.36	-0.57	-0.06	-0.71	-0.72	-0.31
Oil Variance	396.61	766.20	0.51	1.00	0.58	0.09	1.00	0.78	0.35
Equity Variance	105.36	181.59	0.49	0.58	1.00	0.15	0.78	1.00	0.32
Macro Variance	0.48	0.22	0.83	0.09	0.15	1.00	0.35	0.32	1.00

The table reports summary statistics for the macroeconomic, oil sector, asset price, and volatility variables. Means and standard deviations are annualized. Data are quarterly from 1990Q1 to 2019Q4.

(b) Table 1: summary statistics

5.2 Tables 2 and 3: Single and multiple volatility evidence for macroeconomic variables

Read:

- Table 2 shows that oil variance predicts lower future output, consumption, investment, and employment in single-volatility regressions.
- The negative coefficients are often economically larger and statistically stronger than those for equity variance and macro variance.
- Table 3 shows that oil variance remains a strong negative predictor after controlling for equity variance and macro variance in the same regression.
- The oil variance coefficient for investment growth is especially large, implying that uncertainty about oil supply is strongly connected to capital spending.

Economic message: The key message is separability: oil volatility is not driven out by broader uncertainty measures. It is a distinct predictor of real economic activity.

Single volatility evidence: macroeconomic variables.

	Oil Var			Equity Var			Macro Var		
	Slope	SE	Adj. R ²	Slope	SE	Adj. R ²	Slope	SE	Adj. R ²
GDP Growth:									
Current	-0.29	(0.05)	0.41	-0.21	(0.07)	0.30	-0.09	(0.07)	0.20
1q ahead	-0.25	(0.05)	0.20	-0.15	(0.10)	0.09	-0.04	(0.05)	0.03
2q ahead	-0.37	(0.11)	0.15	-0.27	(0.19)	0.09	-0.06	(0.09)	0.02
4q ahead	-0.53	(0.18)	0.08	-0.40	(0.31)	0.04	-0.10	(0.17)	-0.02
12q ahead	-0.47	(0.41)	0.04	-0.26	(0.52)	0.03	0.53	(0.64)	0.05
Consumption Growth:									
Current	-0.15	(0.03)	0.19	-0.15	(0.05)	0.20	-0.08	(0.05)	0.12
1q ahead	-0.15	(0.03)	0.17	-0.12	(0.05)	0.13	-0.07	(0.04)	0.08
2q ahead	-0.24	(0.08)	0.16	-0.18	(0.11)	0.11	-0.12	(0.08)	0.08
4q ahead	-0.43	(0.14)	0.13	-0.34	(0.22)	0.08	-0.21	(0.17)	0.04
12q ahead	-0.52	(0.41)	0.07	-0.40	(0.50)	0.06	-0.16	(0.68)	0.04
Investment Growth:									
Current	-0.82	(0.20)	0.30	-0.47	(0.31)	0.19	-0.31	(0.22)	0.16
1q ahead	-0.96	(0.18)	0.37	-0.62	(0.30)	0.24	-0.12	(0.19)	0.14
2q ahead	-1.60	(0.35)	0.31	-1.15	(0.63)	0.22	-0.13	(0.35)	0.11
4q ahead	-2.74	(0.62)	0.19	-1.95	(1.14)	0.10	0.24	(0.67)	0.00
12q ahead	-1.75	(1.37)	-0.02	-1.16	(1.52)	-0.03	4.73	(1.47)	0.12
Employment Growth:									
Current	-0.24	(0.05)	0.40	-0.20	(0.06)	0.32	-0.12	(0.06)	0.21
1q ahead	-0.25	(0.05)	0.41	-0.22	(0.07)	0.35	-0.07	(0.05)	0.15
2q ahead	-0.47	(0.11)	0.35	-0.43	(0.16)	0.31	-0.10	(0.10)	0.08
4q ahead	-0.80	(0.24)	0.23	-0.76	(0.31)	0.21	-0.03	(0.22)	-0.00
12q ahead	-0.95	(0.46)	0.04	-0.82	(0.47)	0.03	1.27	(0.54)	0.10
TFP Growth:									
Current	0.07	(0.06)	0.02	0.09	(0.05)	0.03	0.01	(0.10)	0.01
1q ahead	0.20	(0.06)	0.09	0.17	(0.07)	0.07	-0.09	(0.06)	0.03
2q ahead	0.38	(0.10)	0.18	0.38	(0.09)	0.18	-0.17	(0.12)	0.08
4q ahead	0.42	(0.19)	0.05	0.60	(0.18)	0.13	-0.44	(0.25)	0.06
12q ahead	0.04	(0.26)	-0.04	0.66	(0.47)	0.01	-0.89	(0.57)	0.05

The table reports the results for the regression of current and future macroeconomic variables on oil, equity, and macroeconomic variance, respectively, controlling for the current and past lags of oil return, oil supply growth, and TFP growth. Newey-West standard errors are in parentheses. Bold numbers indicate statistical significance at the 10-percent level. Data are quarterly from 1990Q1 to 2019Q4.

(a) Table 2: single volatility evidence

Table 3
Multiple volatility evidence: macroeconomic variables.

	Oil Var		Equity Var		Macro Var		Adj. R ²
	Slope	SE	Slope	SE	Slope	SE	
<i>GDP Growth:</i>							
Current	-0.25	(0.04)	-0.08	(0.05)	-0.07	(0.05)	0.43
1q ahead	-0.24	(0.05)	-0.02	(0.10)	-0.03	(0.04)	0.19
2q ahead	-0.31	(0.08)	-0.11	(0.18)	-0.04	(0.08)	0.15
4q ahead	-0.44	(0.12)	-0.16	(0.30)	-0.08	(0.15)	0.07
12q ahead	-0.44	(0.34)	-0.06	(0.46)	0.53	(0.62)	0.05
<i>Consumption Growth:</i>							
Current	-0.09	(0.03)	-0.09	(0.05)	-0.06	(0.04)	0.24
1q ahead	-0.12	(0.03)	-0.05	(0.05)	-0.06	(0.03)	0.19
2q ahead	-0.21	(0.06)	-0.06	(0.10)	-0.11	(0.07)	0.17
4q ahead	-0.35	(0.09)	-0.14	(0.20)	-0.19	(0.14)	0.15
12q ahead	-0.44	(0.33)	-0.16	(0.46)	-0.16	(0.64)	0.06
<i>Investment Growth:</i>							
Current	-0.80	(0.22)	-0.03	(0.30)	-0.27	(0.16)	0.31
1q ahead	-0.87	(0.19)	-0.16	(0.31)	-0.07	(0.12)	0.37
2q ahead	-1.37	(0.33)	-0.43	(0.64)	-0.05	(0.25)	0.31
4q ahead	-2.36	(0.54)	-0.74	(1.12)	0.34	(0.56)	0.19
12q ahead	-1.46	(1.29)	-0.59	(1.10)	4.76	(1.53)	0.12
<i>Employment Growth:</i>							
Current	-0.19	(0.04)	-0.09	(0.05)	-0.10	(0.04)	0.48
1q ahead	-0.18	(0.04)	-0.12	(0.07)	-0.05	(0.03)	0.47
2q ahead	-0.34	(0.08)	-0.24	(0.15)	-0.07	(0.06)	0.41
4q ahead	-0.54	(0.17)	-0.48	(0.29)	0.02	(0.14)	0.28
12q ahead	-0.69	(0.49)	-0.51	(0.39)	1.29	(0.53)	0.16
<i>TFP Growth:</i>							
Current	0.03	(0.06)	0.07	(0.06)	-0.00	(0.10)	0.01
1q ahead	0.15	(0.06)	0.10	(0.08)	-0.10	(0.05)	0.10
2q ahead	0.24	(0.10)	0.28	(0.08)	-0.20	(0.09)	0.25
4q ahead	0.14	(0.16)	0.56	(0.18)	-0.48	(0.21)	0.22
12q ahead	-0.42	(0.31)	0.91	(0.48)	-0.92	(0.55)	0.10

The table reports the results for the regression of current and future macroeconomic variables on oil, equity, and macroeconomic variances, controlling for the current and past lags of oil return, oil supply growth, and TFP growth. Newey-West standard errors are in parentheses. Bold numbers indicate statistical significance at the 10-percent level. Data are quarterly from 1990Q1 to 2019Q4.

future. For example, a one-standard- exhibit a significant negative relation

(b) Table 3: multiple volatility evidence

5.3 Table 4 and Figure 2: Great Recession robustness and rolling-window predictability

Read:

- Table 4 excludes the Great Recession and shows that oil variance still negatively predicts macroeconomic growth.
- Equity variance becomes much less robust outside the crisis window, while oil variance remains negative across macro outcomes.
- Figure 2 shows rolling-window coefficients. Oil variance is generally a stable negative predictor of future GDP, consumption, investment, employment, and oil production.
- TFP responses are weaker and sometimes positive, indicating that the oil-variance channel does not work mainly through aggregate productivity growth.

Economic message: The predictive power of oil variance is not a Great Recession artifact. The rolling estimates show that the relation is persistent across different subperiods.

Table 4
Multiple volatility evidence: excluding great recession.

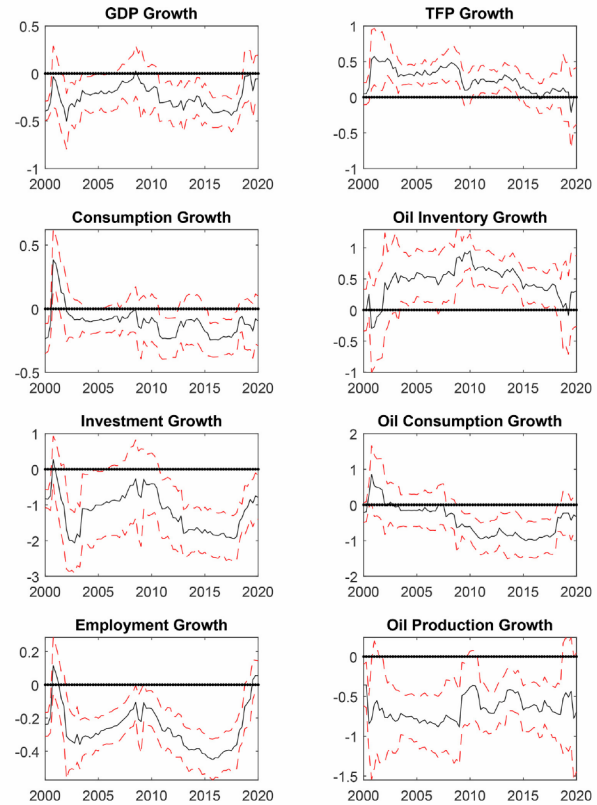
	Oil Var		Equity Var		Macro Var		Adj. R ²
	Slope	SE	Slope	SE	Slope	SE	
GDP Growth:							
Current	-0.16	(0.04)	-0.01	(0.04)	-0.01	(0.03)	0.26
1q ahead	-0.18	(0.05)	0.09	(0.06)	-0.01	(0.03)	0.11
2q ahead	-0.23	(0.09)	0.10	(0.09)	-0.03	(0.07)	0.07
4q ahead	-0.40	(0.10)	0.17	(0.12)	-0.13	(0.12)	0.06
12q ahead	-0.37	(0.20)	0.06	(0.24)	-0.04	(0.30)	-0.01
Consumption Growth:							
Current	-0.07	(0.04)	-0.02	(0.03)	-0.03	(0.03)	0.19
1q ahead	-0.10	(0.03)	0.01	(0.04)	-0.05	(0.03)	0.12
2q ahead	-0.14	(0.05)	0.06	(0.06)	-0.07	(0.05)	0.10
4q ahead	-0.29	(0.08)	0.07	(0.13)	-0.18	(0.12)	0.10
12q ahead	-0.27	(0.19)	0.03	(0.32)	-0.31	(0.38)	0.08
Investment Growth:							
Current	-0.56	(0.15)	0.22	(0.25)	-0.07	(0.13)	0.08
1q ahead	-0.61	(0.13)	0.11	(0.23)	0.01	(0.11)	0.17
2q ahead	-1.07	(0.24)	0.22	(0.38)	0.01	(0.22)	0.22
4q ahead	-2.13	(0.31)	0.49	(0.53)	0.28	(0.43)	0.25
12q ahead	-1.92	(0.85)	-0.64	(1.00)	1.83	(1.06)	0.11
Employment Growth:							
Current	-0.13	(0.03)	-0.03	(0.04)	-0.06	(0.03)	0.21
1q ahead	-0.12	(0.03)	-0.04	(0.04)	-0.02	(0.03)	0.21
2q ahead	-0.22	(0.05)	-0.06	(0.09)	-0.00	(0.06)	0.19
4q ahead	-0.38	(0.08)	-0.10	(0.16)	0.09	(0.12)	0.18
12q ahead	-0.55	(0.23)	-0.24	(0.22)	0.82	(0.26)	0.29
TFP Growth:							
Current	0.04	(0.05)	0.05	(0.05)	-0.04	(0.10)	0.00
1q ahead	0.11	(0.05)	0.07	(0.05)	-0.13	(0.05)	0.08
2q ahead	0.09	(0.07)	0.24	(0.07)	-0.25	(0.09)	0.20
4q ahead	-0.02	(0.09)	0.41	(0.16)	-0.50	(0.22)	0.20
12q ahead	-0.30	(0.25)	1.06	(0.47)	-0.48	(0.59)	0.13
Oil Inventory Growth:							
Current	0.06	(0.11)	-0.24	(0.12)	0.00	(0.09)	0.09
1q ahead	0.17	(0.09)	0.03	(0.08)	0.05	(0.09)	0.00
2q ahead	0.27	(0.13)	0.10	(0.15)	0.09	(0.15)	0.05
4q ahead	0.22	(0.21)	0.20	(0.28)	0.14	(0.30)	0.03
12q ahead	-0.06	(0.55)	0.14	(0.55)	0.27	(0.50)	-0.00
Oil Consumption Growth:							
Current	-0.23	(0.13)	0.23	(0.08)	0.04	(0.07)	0.11
1q ahead	-0.14	(0.11)	-0.14	(0.09)	0.15	(0.07)	0.09
2q ahead	0.03	(0.14)	-0.13	(0.17)	0.39	(0.12)	0.09
4q ahead	0.14	(0.11)	-0.32	(0.21)	0.49	(0.18)	0.11
12q ahead	0.30	(0.28)	-0.48	(0.41)	0.87	(0.35)	0.21
Oil Production Growth:							
Current	-0.07	(0.12)	0.09	(0.10)	-0.03	(0.09)	0.10
1q ahead	-0.20	(0.12)	0.13	(0.09)	0.01	(0.10)	0.13
2q ahead	-0.55	(0.14)	0.29	(0.15)	0.11	(0.14)	0.19
4q ahead	-0.20	(0.15)	0.07	(0.23)	0.22	(0.23)	0.02
12q ahead	-0.57	(0.19)	0.45	(0.46)	0.67	(0.38)	0.08

The table reports the results for the regression of current and future macroeconomic and oil sector variables on oil, equity, and macroeconomic variances, outside of the Great Recession, controlling for the current and past lags of oil return, oil supply growth, and TFP growth. Newey-West standard errors are in parentheses. Bold numbers indicate statistical significance at the 10-percent level. Data are quarterly from 1990Q1

(a) Table 4: excluding the Great Recession

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variation in predictability by oil variance. The figure shows the ten-year rolling window slope coefficients on oil variance in regressions on oil, equity, and macroeconomic variances, controlling for the current and past lags of oil return, oil supply growth, and TFP growth. Dashed line indicates 95% confidence intervals based on Newey-West standard errors. Data are quarterly from 1990Q1 to 2019Q4.

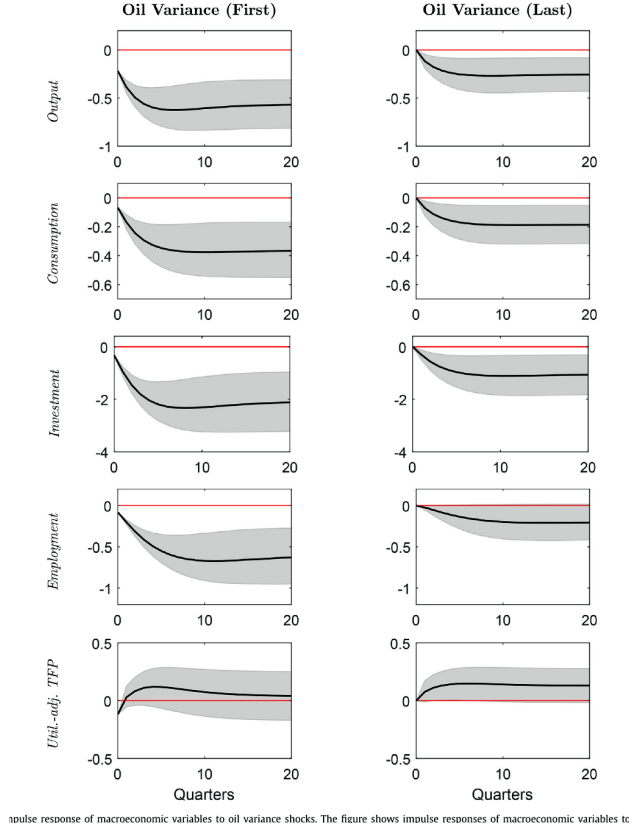
(b) Figure 2: rolling-window slopes

5.4 Figure 3: Impulse response of macroeconomic variables to oil variance shocks

Read:

- A positive oil-variance shock leads to persistent declines in output, consumption, investment, and employment.
- Investment responds most strongly, which is consistent with uncertainty discouraging irreversible or costly-to-adjust capital formation.
- The TFP response is small and not consistently negative, which again suggests that oil variance works through input scarcity, precautionary behavior, and demand rather than pure productivity.
- Results are shown with oil variance ordered first and last in the VAR, reducing dependence on one ordering assumption.

Economic message: The impulse responses translate the regression evidence into dynamic paths: oil uncertainty shocks are contractionary for the real economy.



5.5 Tables 5 and 6; Figure 4: Oil-sector quantities and oil variance shocks

Read:

- Tables 5 and 6 show that oil variance predicts higher oil inventory growth and lower oil consumption and production growth.
- The oil inventory result is crucial because it supports precautionary storage rather than a pure demand-collapse interpretation.
- Figure 4 confirms the dynamic pattern: inventories rise after an oil-variance shock, while oil consumption and oil production tend to decline.
- The responses are robust to the VAR ordering of oil variance.

Economic message: Higher oil uncertainty makes agents hold more oil inventories as insurance. This storage response tightens effective oil supply for current production, transmitting uncertainty into the macroeconomy.

Table 5
Single volatility evidence: oil sector variables.

	Oil Var			Equity Var			Macro Var		
	Slope	SE	Adj. R ²	Slope	SE	Adj. R ²	Slope	SE	Adj. R ²
<i>Oil Inventory Growth:</i>									
Current	0.01	(0.08)	0.12	-0.11	(0.13)	0.13	0.04	(0.08)	0.12
1q ahead	0.24	(0.09)	0.05	0.17	(0.10)	0.02	0.04	(0.09)	0.00
2q ahead	0.34	(0.15)	0.07	0.33	(0.19)	0.07	0.03	(0.14)	0.03
4q ahead	0.34	(0.23)	0.05	0.45	(0.30)	0.06	0.10	(0.27)	0.03
12q ahead	-0.18	(0.44)	-0.02	-0.12	(0.42)	-0.02	-0.28	(0.49)	-0.02
<i>Oil Consumption Growth:</i>									
Current	-0.19	(0.12)	0.07	-0.10	(0.15)	0.05	-0.01	(0.09)	0.04
1q ahead	-0.37	(0.10)	0.14	-0.31	(0.10)	0.11	0.08	(0.08)	0.05
2q ahead	-0.33	(0.20)	0.05	-0.38	(0.17)	0.07	0.24	(0.15)	0.03
4q ahead	-0.38	(0.27)	0.04	-0.64	(0.24)	0.11	0.45	(0.21)	0.06
12q ahead	-0.34	(0.57)	-0.00	-0.43	(0.66)	0.00	1.35	(0.88)	0.12
<i>Oil Production Growth:</i>									
Current	-0.91	(0.21)	0.29	-0.57	(0.34)	0.15	-0.36	(0.24)	0.10
1q ahead	-0.96	(0.18)	0.37	-0.62	(0.30)	0.24	-0.12	(0.19)	0.14
2q ahead	-1.60	(0.35)	0.31	-1.15	(0.63)	0.22	-0.13	(0.35)	0.11
4q ahead	-2.74	(0.62)	0.19	-1.95	(1.14)	0.10	0.24	(0.67)	0.00
12q ahead	-1.75	(1.37)	-0.02	-1.16	(1.52)	-0.03	4.73	(1.47)	0.12

The table reports the results for the regression of current and future oil sector variables on oil, equity, and macroeconomic variance, respectively, controlling for the current and past lags of oil return, oil supply growth, and TFP growth. Newey-West standard errors are in parentheses. Bold numbers indicate statistical significance at the 10-percent level. Data are quarterly from 1990Q1 to 2019Q4.

(a) Table 5: single volatility, oil sector

Table 6
Multiple volatility evidence: oil sector variables.

	Oil Var		Equity Var		Macro Var		Adj. R ²
	Slope	SE	Slope	SE	Slope	SE	
<i>Oil Inventory Growth:</i>							
Current	0.10	(0.11)	-0.17	(0.15)	0.05	(0.09)	0.12
1q ahead	0.21	(0.09)	0.05	(0.11)	0.03	(0.09)	0.03
2q ahead	0.22	(0.15)	0.22	(0.20)	0.01	(0.15)	0.06
4q ahead	0.14	(0.22)	0.37	(0.31)	0.07	(0.30)	0.04
12q ahead	-0.16	(0.51)	-0.02	(0.50)	-0.28	(0.50)	-0.03
<i>Oil Consumption Growth:</i>							
Current	-0.19	(0.17)	0.00	(0.19)	0.00	(0.09)	0.05
1q ahead	-0.28	(0.12)	-0.17	(0.11)	0.10	(0.07)	0.15
2q ahead	-0.17	(0.23)	-0.32	(0.21)	0.27	(0.13)	0.10
4q ahead	-0.04	(0.26)	-0.65	(0.27)	0.49	(0.21)	0.16
12q ahead	-0.13	(0.64)	-0.41	(0.63)	1.36	(0.84)	0.11
<i>Oil Production Growth:</i>							
Current	-0.16	(0.13)	0.04	(0.12)	-0.06	(0.09)	0.10
1q ahead	-0.22	(0.11)	0.12	(0.10)	0.00	(0.10)	0.15
2q ahead	-0.53	(0.15)	0.22	(0.18)	0.13	(0.14)	0.18
4q ahead	-0.12	(0.15)	-0.06	(0.25)	0.29	(0.23)	-0.01
12q ahead	-0.72	(0.17)	0.40	(0.46)	0.82	(0.42)	0.10

The table reports the results for the regression of current and future oil sector variables by oil, equity, and macroeconomic variances, controlling for the current and past lags of oil return, oil supply growth, and TFP growth. Newey-West standard errors are in parentheses. Bold numbers indicate statistical significance at the 10-percent level. Data are quarterly from 1990Q1 to 2019Q4.

es between -0.15% and -0.24%, and it any horizons or for either ordering. Online Appendix Figs. 1A.5 and 1A.6

effect is mostly insignificant. Equity an variances do not have an appreciable im tories or oil production.

(b) Table 6: multiple volatility, oil sector

5.6 Figure 5: Volatility exposures of oil basis

Read:

- The left panels show a negative relationship between oil basis and oil inventory growth, consistent with classical storage theory.
- The right panels show that the oil basis has a strong negative exposure to oil variance, especially at longer horizons.
- Exposures to equity variance and macro variance are weaker and often positive.
- The negative oil-variance exposure of the oil basis connects option-implied uncertainty to spot-futures pricing.

Economic message: Oil variance is priced in the futures market through inventories and storage. The oil basis evidence provides market-price support for the precautionary-inventory mechanism.

5.7 Figure 6 and Table 7: Portfolio exposures and sectoral dividends

Read:

- Figure 6 shows that the oil sector has positive exposure to oil variance and oil price risk, while many non-oil sectors and the aggregate market have negative oil-variance exposures.
- The cross-section shows a tradeoff: industries that benefit from high oil prices or oil scarcity are not the same as industries harmed by oil uncertainty.
- Table 7 shows that oil variance is negatively associated with future dividends for many sectors, especially the lowest and medium terciles of oil-price beta.
- The oil sector is less harmed because its cash flows benefit from higher scarcity and oil-sector valuation effects.

Economic message: Oil uncertainty is a priced cross-sectional risk. It depresses broad market and non-oil cash-flow expectations, but it can raise the relative value of oil-sector claims.

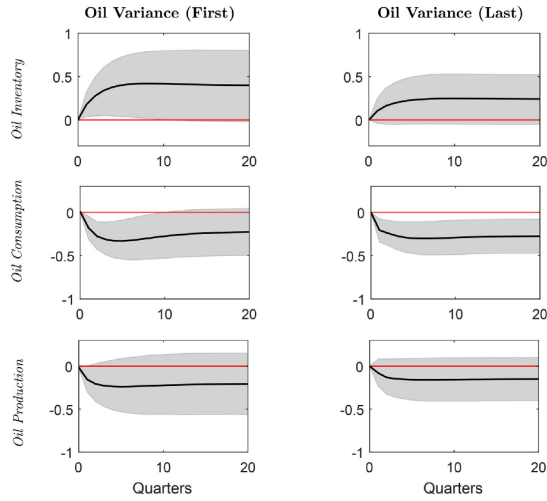


Figure 6: Portfolio exposures. The figure shows impulse responses of oil sector variables to a positive oil variance shock. The impulse responses are based on a lower-triangular Cholesky decomposition of a VAR(1) fit to the following series: oil return, oil supply growth, TFP growth, and volatility measures (oil, equity, and macroeconomic variances). In the left (right) column, the response is ordered first (last) among the variables. Gray regions indicate 90% confidence intervals computed by block bootstrap. Data are quarterly from 1990:1 to 2019:4. Changes are in percent.

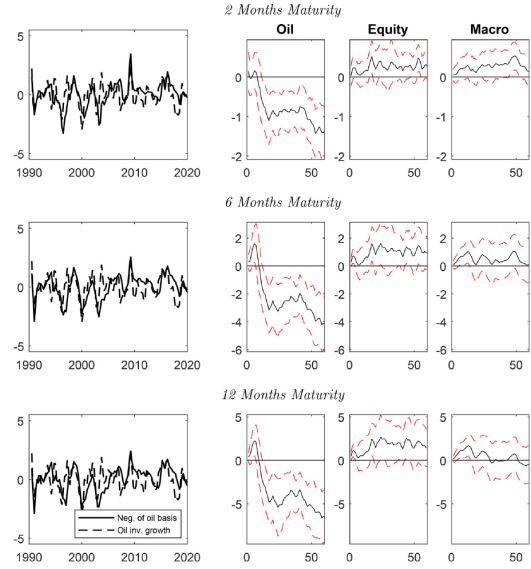


Figure 7: Volatility exposures of oil basis. The left panels of the figure show the time series of negative oil basis (solid line) and oil inventory growth (dashed line), standardized to have mean zero and variance one. The right panels show the exposures of the oil basis to oil, equity, and macroeconomic variances at horizons of one month to five years. The top panel shows the evidence for oil contracts with two months to maturity, the middle panel for six months, and the bottom panel for 12 months. Data are monthly from 1990M1 to 2019M12.

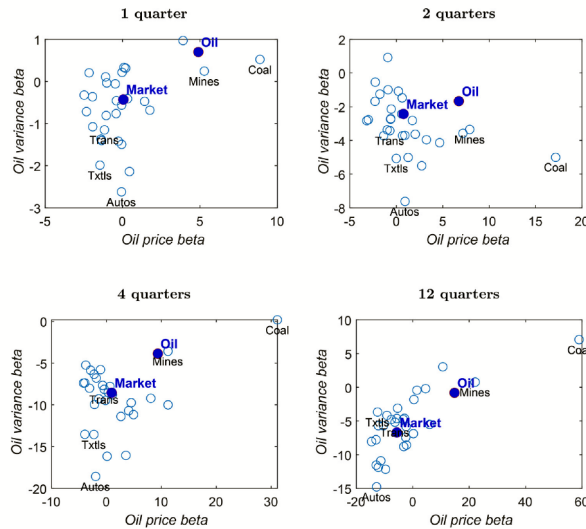


Figure 8: Portfolio exposures. The figure shows the exposures of the aggregate market and the 30 Fama-French industry portfolios to oil variance and oil price risk. The figure shows the exposures of the aggregate market and the 30 Fama-French industry portfolios to oil variance and oil price risk. The figure shows the exposures of the aggregate market and the 30 Fama-French industry portfolios to oil variance and oil price risk.

(a) Figure 6: portfolio exposures

Table 7
Multiple volatility evidence: sectoral dividends.

	Oil Var		Equity Var		Macro Var		Adj. R ²
	Slope	SE	Slope	SE	Slope	SE	
Aggregate Market:							
Current	-0.10	(0.24)	-0.38	(0.21)	-0.69	(0.28)	0.09
1q ahead	-0.11	(0.27)	-0.46	(0.33)	-0.59	(0.25)	0.06
2q ahead	-0.79	(0.44)	-0.60	(0.58)	-1.12	(0.45)	0.14
4q ahead	-1.76	(0.75)	-0.80	(1.27)	-1.42	(0.67)	0.17
12q ahead	-1.61	(1.85)	1.42	(2.72)	2.68	(2.58)	-0.01
Lowest Tercile:							
Current	-0.21	(0.38)	-0.83	(0.49)	-0.60	(0.42)	0.01
1q ahead	-0.56	(0.35)	-0.61	(0.42)	-0.57	(0.40)	0.02
2q ahead	-1.15	(0.62)	-0.70	(0.87)	-0.91	(0.74)	0.08
4q ahead	-2.22	(1.07)	-0.42	(2.00)	-0.55	(1.30)	0.11
12q ahead	-4.21	(3.24)	6.72	(3.85)	7.64	(4.10)	0.06
Medium Tercile:							
Current	0.39	(0.35)	-0.46	(0.28)	-0.17	(0.27)	0.03
1q ahead	-0.06	(0.32)	-0.54	(0.40)	-0.23	(0.24)	0.01
2q ahead	-0.67	(0.40)	-0.53	(0.51)	-0.46	(0.44)	0.04
4q ahead	-1.53	(0.59)	-0.49	(0.68)	-0.59	(0.76)	0.10
12q ahead	-1.27	(1.83)	1.49	(2.34)	0.66	(2.62)	-0.06
Highest Tercile:							
Current	-0.12	(0.44)	-0.22	(0.44)	-0.92	(0.46)	-0.02
1q ahead	0.21	(0.40)	-0.65	(0.55)	-0.48	(0.49)	-0.04
2q ahead	0.04	(0.79)	-1.12	(1.14)	-0.63	(1.04)	-0.03
4q ahead	-0.34	(1.38)	-1.03	(1.93)	-0.32	(1.92)	0.03
12q ahead	3.62	(5.52)	3.27	(3.15)	1.98	(2.96)	0.01

The table reports the results for the regression of current and future sectoral dividends on oil, equity, and macroeconomic variances, controlling for the current and past lags of oil return, oil supply growth, and TFP growth. The 30 Fama-French industry portfolios are sorted into terciles based on the oil variance betas of their equity returns at four-quarter frequency. Newey-West standard errors are in parentheses. Bold numbers indicate statistical significance at the 10-percent level.

(b) Table 7: sectoral dividends

5.8 Table 8: Volatility and macroeconomic forecasts

Read:

- Oil variance predicts weaker Survey of Professional Forecasters growth forecasts for real GDP, consumption, investment, industrial production, and profits.
- Oil variance is also positively related to uncertainty indexes, especially macro and financial uncertainty.
- Equity variance is more closely connected to financial uncertainty, while oil variance has a broader real-activity forecast component.

Economic message: Expectations data confirm that oil variance is not just realized macro weakness. It is associated with lower expected future real activity and higher perceived uncertainty.

Table 8
Volatility and macroeconomic forecasts.

	Oil Var		Equity Var		Macro Var		Adj. R ²
	Slope	SE	Slope	SE	Slope	SE	
Growth Forecasts							
<i>Real GDP:</i>							
1q forecast	-0.17	(0.03)	-0.06	(0.04)	-0.01	(0.02)	0.51
4q forecast	-0.08	(0.02)	-0.03	(0.03)	0.01	(0.01)	0.41
<i>Real Consumption:</i>							
1q forecast	-0.14	(0.02)	-0.05	(0.04)	-0.03	(0.02)	0.41
4q forecast	-0.06	(0.01)	-0.03	(0.02)	-0.02	(0.01)	0.37
<i>Nonresidential Investment:</i>							
1q forecast	-0.56	(0.10)	-0.14	(0.17)	-0.17	(0.10)	0.46
4q forecast	-0.37	(0.06)	-0.10	(0.11)	-0.07	(0.06)	0.49
<i>Industrial Production:</i>							
1q forecast	-0.39	(0.04)	-0.08	(0.07)	0.06	(0.04)	0.59
4q forecast	-0.16	(0.02)	-0.05	(0.04)	0.08	(0.02)	0.53
<i>Real Corporate Profits:</i>							
1q forecast	-0.49	(0.26)	-0.17	(0.23)	0.44	(0.16)	0.15
4q forecast	-0.18	(0.11)	-0.08	(0.10)	0.37	(0.09)	0.19
Uncertainty Indices							
<i>Real Uncertainty:</i>							
1q uncertainty	1.84	(0.60)	0.72	(0.64)	0.08	(0.49)	0.28
4q uncertainty	1.40	(0.42)	0.54	(0.48)	-0.13	(0.31)	0.31
<i>Macro Uncertainty:</i>							
1q uncertainty	4.06	(1.41)	2.23	(1.30)	0.67	(0.84)	0.43
4q uncertainty	2.77	(0.78)	1.41	(0.82)	0.20	(0.43)	0.47
<i>Fin Uncertainty:</i>							
1q uncertainty	2.56	(1.17)	9.45	(1.33)	-0.65	(1.01)	0.60
4q uncertainty	0.90	(0.45)	3.42	(0.51)	-0.28	(0.37)	0.58

The table reports the results for the regression of current growth forecasts and uncertainty indicators on oil, equity, and macroeconomic variances, controlling for the current and past lags of oil return.

5.9 Tables 9 and 10: Model calibration and model moments

Read:

- Table 9 reports preference, macro, and oil-sector parameters for the baseline and amplified model.
- Key oil-sector parameters govern the oil input share, substitutability between oil and capital, inventory costs, and stochastic oil productivity and volatility.
- Table 10 shows that the calibrated model matches the oil input share, the inventory-consumption ratio, and important oil price variance moments.
- The model also produces realistic correlations between oil returns and oil variance and broadly plausible investment-output and consumption-output volatility ratios.

Economic message: The calibration is designed to discipline the mechanism. The model must reproduce oil-market moments before it is used to explain macro and asset-price responses to oil uncertainty.

Table 9
Model parameters.

Parameter	Value	
	Baseline	Amplified
<i>Preferences</i>		
Subjective discount factor	β	0.96
Risk aversion	γ	10
Intertemporal elasticity of substitution	ψ	2
<i>General Macroeconomy</i>		
Capital share	α	0.34
Depreciation rate of capital	δ	6%
Average growth rate	μ	1.8%
Capital adjustment costs	ξ	3.5 3.0
Share of final goods in consumption	τ	0.205
Elasticity of substitution between leisure and consumption goods	ξ_l	1.2
Volatility of short-run macro productivity risk	σ_A	3.35%
Autocorrelation of expected growth	ϕ	0.98
Volatility of long-run macro productivity risk	σ_x	0.1 σ_A
Mean-reversion of volatility of macro productivity	ρ_w	0.962
Volatility of volatility of macro productivity	σ_w	3.46 %
Price markup elasticity	ε_θ	— -0.1
Share of intermediate goods sector	ν	— 0.5
<i>Oil Sector</i>		
Elasticity of substitution between oil and capital	ϕ	0.400 0.225
Oil intensity parameter ($\bar{l} = l^{\frac{1}{2}}$)	l	0.011 0.6
Oil inventory costs	ω	10%
Oil stock-out costs	π	$2.5 \cdot 10^{-7}$ $1.5 \cdot 10^{-7}$
Average oil production rate	η	8%
Mean-reversion of oil productivity	χ	0.966
Volatility of oil productivity	σ_κ	16.2% 14.8%
Mean-reversion of oil production volatility	ρ_v	0.58 0.28
Volatility of oil production volatility	σ_v	57% 41%

The table reports the parameters of the calibrated model, both for the baseline and the amplified version. Parameters describing the household's preferences as well as the general structure of the macroeconomy and the oil sector are set according to the literature. The oil intensity parameter, the oil stock-out cost

(a) Table 9: model parameters

Table 10
Moments.

Statistic	Quantity Moments Data	Model	
		Baseline	Amplified
<i>Quantity Moments</i>			
Oil input relative to general consumption			
$\mathbb{E}[P^*/J/C]$	0.04	0.04	0.04
Oil inventory-consumption ratio			
$\mathbb{E}[S/J]$	0.25	0.25	0.25
Relative volatility of oil inventories and oil consumption			
$\sigma(\Delta S)/\sigma(\Delta J)$	0.96	0.43	1.13
Investment-output ratio			
$\mathbb{E}[I/Y]$ [%]	16.01	27.21	20.14
Relative volatility of general investment and output			
$\sigma(\Delta I)/\sigma(\Delta Y)$	3.23	2.73	2.41
Relative volatility of general consumption and output			
$\sigma(\Delta C)/\sigma(\Delta Y)$	0.73	0.61	0.71
<i>Price Moments</i>			
Statistic	Data	Model	
		Baseline	Amplified
Mean of oil price variance			
$\mathbb{E}[\sigma_p^2(\Delta p)]$	396.61	398.43	396.22
Volatility of oil price variance			
$\sigma(\sigma_p^2(\Delta p))$	766.20	757.31	754.70
Autocorrelation of oil price variance			
$\rho(\sigma_{t-1}^2(\Delta p), \sigma_t^2(\Delta p))$	0.51	0.51	0.51
Correlation between oil return and oil variance			
$\rho(\Delta p, \sigma_p^2(\Delta p))$	-0.27	0.13	0.11
Risk-free rate			
$\mathbb{E}[r_t^f]$ [%]	0.34	1.58	1.54
Volatility of risk-free rate			
$\sigma(r_t^f)$ [%]	0.81	0.72	0.91
Equity risk premium			
$\mathbb{E}[r_{EX,t+1}^{LEV}]$ [%]	6.49	3.27	2.78

The table reports the quantity and price moments produced by our model, both for the baseline and the amplified version, compared to their empirical counterparts. We simulate

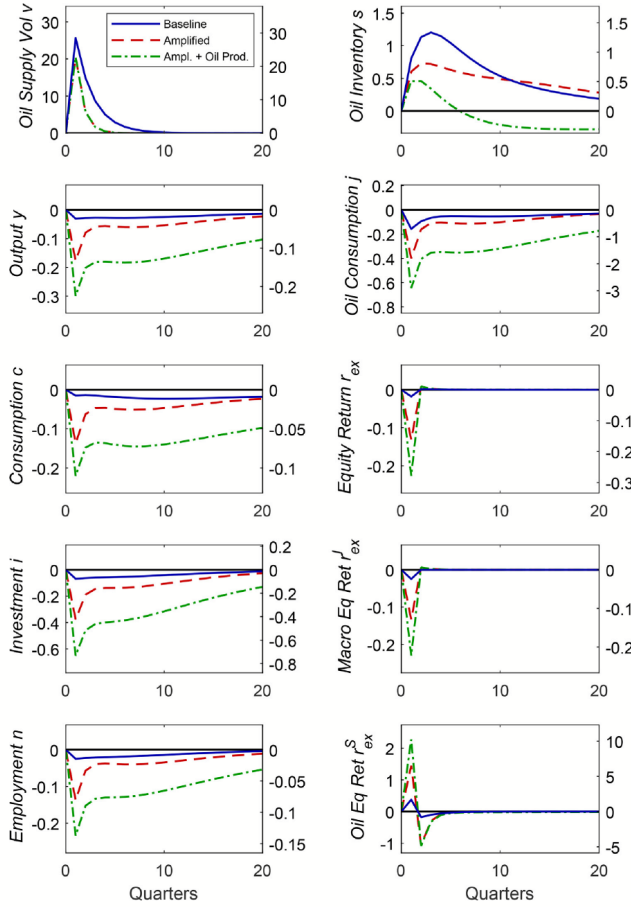
(b) Table 10: model moments

5.10 Figures 7 and 8: Model-based impulse responses to oil uncertainty shocks

Read:

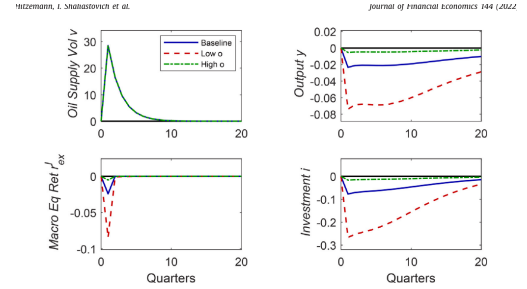
- Figure 7 shows that a positive oil uncertainty shock increases oil supply volatility and inventories while reducing output, consumption, investment, employment, and aggregate equity returns.
- The amplified model generates stronger responses, bringing model dynamics closer to the empirical impulse responses.
- Oil equity returns move differently from macro equity returns, consistent with the empirical cross-sectional asset-pricing evidence.
- Figure 8 shows that the contractionary macro effect is larger when oil and capital are harder to substitute.
- When oil input sensitivity is high, output and investment decline more sharply after oil uncertainty shocks.

Economic message: The model formalizes the mechanism: oil uncertainty raises precautionary storage, reduces effective oil input, and depresses macro activity. Substitution possibilities determine the amplification.



Model-based impulse responses to oil uncertainty shocks. The figure shows model-based impulse response functions for a positive one-standard-deviation shock to oil production volatility v_t . The blue solid lines and the right axis stand for the baseline model, the left axis and red dashed lines for the amplified model with time-varying markups, and green dot-dashed lines for the amplified model with time-varying markups and a negative

(a) Figure 7: model impulse responses



Model-based impulse responses to oil uncertainty shocks: different oil input sensitivities. The figure shows model-based impulse response functions for a positive one-standard-deviation shock to oil production volatility v_t . The impulse responses are calculated for the baseline model calibration with $\alpha = 0.4$. The blue solid lines stand for the baseline calibration with $\alpha = 0.4$, the red dashed lines for $\alpha = 0.35$, and the green dot-dashed lines for $\alpha = 0.45$. Changes are in percent.

(b) Figure 8: oil input sensitivities

5.11 Tables 11, 12, and 13: Model-based predictive regressions

Read:

- Table 11 shows that simulated oil variance predicts weaker macro activity, stronger oil inventories, lower oil consumption, and positive oil equity returns in single-volatility regressions.
- Table 12 shows that the model can reproduce the dominance of oil variance in multiple-volatility regressions.
- Table 13 demonstrates that the result depends on distinguishing fundamental and non-fundamental components of uncertainty. If equity variance is made too fundamentally informative, it can dominate, but realistic cases preserve the oil-variance channel.
- Across simulations, oil variance remains a robust indicator of macroeconomic activity when the model features the paper's precautionary oil-storage mechanism.

Economic message: The model is not only qualitatively intuitive. It reproduces the empirical regression structure: oil variance stands out as a predictor of future macro and oil-sector conditions.

Table 11
Single volatility evidence: model simulations.

	Amplified Model, $\text{corr}(v^o, v^e) = 0.0$				Macro Var				
	Oil Var		Equity Var		Slope		SigRatio		
	Slope	SigRatio	Adj. R ²	Slope	SigRatio	Adj. R ²	Slope	SigRatio	
Output growth	-0.20	[0.72]	0.91	-0.13	[0.93]	0.91	0.02	[0.13]	0.91
Consumption growth	-0.04	[0.72]	0.90	-0.13	[0.90]	0.89	-0.06	[0.16]	0.90
Investment growth	-0.15	[0.72]	0.84	-0.22	[0.72]	0.84	0.39	[0.26]	0.84
Employment growth	-0.10	[0.72]	0.90	-0.22	[0.72]	0.87	0.02	[0.07]	0.90
TFP growth	0.06	[0.10]	-0.00	0.05	[0.11]	-0.00	-0.02	[0.11]	0.01
Oil inventory growth	0.51	[0.95]	0.95	0.33	[0.98]	0.95	0.04	[0.11]	0.95
Oil consumption growth	-0.40	[0.93]	0.95	-0.26	[0.94]	0.95	-0.01	[0.13]	0.95
Aggregate market return	-0.21	[0.71]	0.84	-0.17	[0.64]	0.84	0.15	[0.20]	0.84
Non-oil equity return	-0.20	[0.71]	0.85	-0.17	[0.64]	0.84	0.15	[0.20]	0.84
Oil equity return	1.52	[1.00]	0.95	1.11	[0.94]	0.95	0.22	[0.11]	0.95
	Amplified Model, $\text{corr}(v^o, v^e) = 0.0$ without non-fundamental oil variance shocks v^o				Macro Var				
	Oil Var		Equity Var		Slope		SigRatio		
	Slope	SigRatio	Adj. R ²	Slope	SigRatio	Adj. R ²	Slope	SigRatio	
Output growth	-0.30	[0.90]	0.92	-0.13	[0.93]	0.91	0.02	[0.13]	0.91
Consumption growth	-0.23	[0.90]	0.90	-0.13	[0.90]	0.89	-0.06	[0.16]	0.90
Investment growth	-0.09	[0.93]	0.85	-0.22	[0.72]	0.84	0.39	[0.26]	0.84
Employment growth	-0.10	[0.93]	0.88	-0.04	[0.91]	0.87	0.02	[0.07]	0.88
TFP growth	0.06	[0.10]	0.00	0.05	[0.11]	-0.00	-0.02	[0.11]	0.01
Oil inventory growth	0.81	[1.00]	0.96	0.31	[0.98]	0.95	0.04	[0.11]	0.95
Oil consumption growth	-0.65	[0.99]	0.95	-0.28	[0.94]	0.95	-0.01	[0.13]	0.95
Aggregate market return	-0.27	[0.70]	0.84	-0.16	[0.73]	0.84	0.15	[0.30]	0.84
Non-oil equity return	-0.27	[0.70]	0.85	-0.17	[0.64]	0.84	0.15	[0.20]	0.84
Oil equity return	2.75	[1.00]	0.95	1.11	[0.94]	0.95	0.22	[0.11]	0.95
	Amplified Model, $\text{corr}(v^o, v^e) = -0.25$				Macro Var				
	Oil Var		Equity Var		Slope		SigRatio		
	Slope	SigRatio	Adj. R ²	Slope	SigRatio	Adj. R ²	Slope	SigRatio	
Output growth	-0.19	[0.60]	0.92	-0.13	[0.93]	0.91	-0.12	[0.27]	0.92
Consumption growth	-0.18	[0.72]	0.89	-0.12	[0.91]	0.89	-0.12	[0.33]	0.88
Investment growth	-0.31	[0.70]	0.84	-0.22	[0.72]	0.85	-0.08	[0.09]	0.85
Employment growth	-0.10	[0.28]	0.88	-0.07	[0.22]	0.88	-0.35	[0.47]	0.89
TFP growth	0.06	[0.20]	0.01	0.04	[0.15]	0.00	-0.46	[0.62]	0.04
Oil inventory growth	0.54	[0.94]	0.95	0.30	[0.98]	0.95	0.05	[0.11]	0.95
Oil consumption growth	-0.44	[0.97]	0.95	-0.26	[0.92]	0.95	-0.02	[0.12]	0.95
Aggregate market return	-0.19	[0.61]	0.85	-0.15	[0.51]	0.85	0.00	[0.12]	0.85
Non-oil equity return	-0.19	[0.60]	0.86	-0.15	[0.67]	0.86	0.00	[0.11]	0.85
Oil equity return	1.91	[0.95]	0.96	1.25	[0.92]	0.96	0.10	[0.11]	0.95

The table reports the results for regressions of macroeconomic, oil sector, and asset price variables on oil, equity, and macroeconomic variances, respectively, based on simulated data from our model. Controls include the contemporaneous growth rate of oil production as well as previous growth rates of oil production and TFP, the variable of interest, the oil return, and the considered uncertainty variable. Slope coefficients and Newey-West standard errors are estimated for 100 simulated economies of 120 quarters length. We report the median slope coefficients across these economies, the median adjusted R², and in square brackets the ratio of economies for which the estimated coefficient is statistically significant at the 10% level. Bold numbers indicate significance for at least 50 of the 100 sample economies.

Table 12
Multiple volatility evidence: model simulations.

	Amplified Model, $\text{corr}(v^o, v^e) = 0.0$				Macro Var			
	Oil Var		Equity Var		Slope		SigRatio	
	Slope	SigRatio	Slope	SigRatio	Slope	SigRatio	Slope	SigRatio
Output growth	-0.17 [0.90]		-0.09	[0.20]	0.01	[0.12]	0.91	
Consumption growth	-0.15 [0.90]		-0.08	[0.31]	-0.05	[0.18]	0.90	
Investment growth	-0.29	[0.19]	-0.16	[0.14]	0.39	[0.30]	0.84	
Employment growth	-0.09	[0.25]	-0.05	[0.14]	0.02	[0.10]	0.88	
TFP growth	0.05	[0.19]	0.02	[0.10]	-0.03	[0.12]	0.91	
Oil inventory growth	0.46 [0.91]		0.17	[0.36]	0.02	[0.08]	0.95	
Oil consumption growth	-0.37 [0.75]		-0.16	[0.36]	-0.01	[0.12]	0.95	
Aggregate market return	-0.18	[0.47]	-0.07	[0.18]	0.14	[0.33]	0.84	
Non-oil equity return	-0.18	[0.47]	-0.07	[0.18]	0.14	[0.33]	0.85	
Oil equity return	1.69 [0.95]		0.56	[0.42]	0.19	[0.11]	0.96	
	Amplified Model, $\text{corr}(v^o, v^e) = 0.0$, without non-fundamental oil variance shocks v^o				Macro Var			
	Oil Var		Equity Var		Slope		SigRatio	
	Slope	SigRatio	Slope	SigRatio	Slope	SigRatio	Slope	SigRatio
Output growth	-0.27 [0.75]		-0.04	[0.22]	0.03	[0.10]	0.92	
Consumption growth	-0.24 [0.75]		-0.05	[0.28]	-0.04	[0.16]	0.90	
Investment growth	-0.57	[0.42]	-0.10	[0.15]	0.39	[0.30]	0.85	
Employment growth	-0.20	[0.27]	-0.03	[0.17]	0.03	[0.10]	0.88	
TFP growth	0.06	[0.17]	0.02	[0.15]	-0.03	[0.11]	0.91	
Oil inventory growth	0.74 [0.96]		0.07	[0.20]	0.01	[0.08]	0.96	
Oil consumption growth	-0.60 [0.90]		-0.06	[0.26]	-0.01	[0.15]	0.95	
Aggregate market return	-0.26 [0.61]		-0.04	[0.17]	0.14	[0.34]	0.84	
Non-oil equity return	-0.26 [0.64]		-0.04	[0.17]	0.14	[0.33]	0.85	
Oil equity return	2.55 [0.99]		0.21	[0.27]	0.18	[0.12]	0.96	
	Amplified Model, $\text{corr}(v^o, v^e) = -0.25$				Macro Var			
	Oil Var		Equity Var		Slope		SigRatio	
	Slope	SigRatio	Slope	SigRatio	Slope	SigRatio	Slope	SigRatio
Output growth	-0.17 [0.51]		-0.09	[0.25]	-0.14	[0.28]	0.92	
Consumption growth	-0.14 [0.57]		-0.08	[0.34]	-0.11	[0.34]	0.90	
Investment growth	-0.29	[0.34]	-0.17	[0.15]	-0.01	[0.07]	0.86	
Employment growth	-0.07	[0.21]	-0.06	[0.16]	-0.34	[0.49]	0.89	
TFP growth	0.04	[0.12]	0.01	[0.13]	-0.45 [0.50]	0.04		
Oil inventory growth	0.68 [0.96]		0.14	[0.29]	0.00	[0.09]	0.96	
Oil consumption growth	-0.35 [0.78]		-0.17	[0.36]	-0.00	[0.13]	0.96	
Aggregate market return	-0.16	[0.40]	-0.07	[0.20]	0.00	[0.09]	0.85	
Non-oil equity return	-0.16	[0.44]	-0.07	[0.20]	0.00	[0.09]	0.86	
Oil equity return	1.56 [0.92]		0.59	[0.46]	0.05	[0.09]	0.96	

The table reports the results for regressions of macroeconomic, oil sector, and asset price variables on oil, equity, and macroeconomic variances based on simulated data from our model. Controls include the contemporaneous growth rate of oil production as well as previous growth rates of oil production and TFP, the variable of interest, the oil return, and the considered uncertainty variable. Slope coefficients and Newey-West standard errors are estimated for 100 simulated economies of 120 quarters length. We report the median slope coefficients across these economies, the median adjusted R², and in square brackets the ratio of economies for which the estimated coefficient is statistically significant at the 10% level. Bold numbers indicate significance for at least 50 of the 100 sample economies (i.e., for the median simulated economy). The same word indicates the corresponding rate that another variance is not robust to

Table 13
Multiple volatility evidence: model simulations, additional cases.

	Amplified Model, without non-fundamental equity variance shocks v^e				Macro Var			
	Oil Var		Equity Var		Slope		SigRatio	
	Slope	SigRatio	Slope	SigRatio	Slope	SigRatio	Slope	SigRatio
Output growth	0.13	[0.29]	-0.56 [0.89]	0.03	[0.16]	0.92		
Consumption growth	0.13	[0.40]	-0.50 [0.93]	-0.04	[0.15]	0.91		
Investment growth	0.33	[0.21]	1.21 [0.62]	0.42	[0.26]	0.85		
Employment growth	0.11	[0.22]	-0.33	[0.37]	0.03	[0.12]	0.69	
TFP growth	0.02	[0.14]	0.04	[0.15]	-0.04	[0.11]	0.91	
Oil inventory growth	-0.17	[0.37]	1.19	[0.99]	0.01	[0.00]	0.96	
Oil consumption growth	0.23	[0.34]	-1.90 [0.99]	0.00	[0.15]	0.96		
Aggregate market return	0.06	[0.22]	-0.48 [0.66]	0.15	[0.37]	0.85		
Non-oil equity return	0.09	[0.23]	-0.48 [0.68]	0.15	[0.37]	0.85		
Oil equity return	-0.56	[0.25]	1.78	[0.00]	0.12	[0.11]	0.96	
	Amplified Model, $\text{corr}(v^o, v^e) = -0.7$				Macro Var			
	Oil Var		Equity Var		Slope		SigRatio	
	Slope	SigRatio	Slope	SigRatio	Slope	SigRatio	Slope	SigRatio
Output growth	-0.16 [0.52]		-0.09	[0.25]	-0.27 [0.76]	0.92		
Consumption growth	-0.15 [0.57]		-0.08	[0.33]	-0.19 [0.67]	0.89		
Investment growth	-0.27	[0.21]	-0.15	[0.18]	-0.48	[0.45]	0.87	
Employment growth	-0.08	[0.20]	-0.08	[0.16]	-0.64 [0.95]	0.72		
TFP growth	0.07	[0.14]	-0.02	[0.14]	-0.85 [0.97]	0.13		
Oil inventory growth	0.65 [0.78]		0.17	[0.37]	0.06	[0.16]	0.95	
Oil consumption growth	-0.36 [0.75]		-0.16	[0.36]	-0.01	[0.14]	0.96	
Aggregate market return	-0.16	[0.48]	-0.07	[0.17]	-0.14	[0.33]	0.87	
Non-oil equity return	-0.16	[0.48]	-0.07	[0.18]	-0.14	[0.33]	0.87	
Oil equity return	1.66 [0.90]		0.59	[0.45]	-0.05	[0.16]	0.96	

The table reports the results for multivariate regression of macroeconomic, oil sector, and asset price variables on oil, equity, and macroeconomic variances based on simulated data from our model. Controls include the contemporaneous growth rate of oil production as well as previous growth rates of oil production and TFP, the variable of interest, the oil return, and the considered uncertainty variable. Slope coefficients and Newey-West standard errors are estimated for 100 simulated economies of 120 quarters length. We report the median slope coefficients across these economies, the median adjusted R², and in square brackets the ratio of economies for which the estimated coefficient is statistically significant at the 10% level. Bold numbers indicate significance for at least 50 of the 100 sample economies (i.e., for the median simulated economy). The same word indicates the corresponding rate that another variance is not robust to

6 Short summary

- The paper studies option-implied oil variance as a macro-financial risk factor.
- Oil variance predicts lower future GDP, consumption, investment, and employment growth.
- The predictive power remains after controlling for equity variance and macroeconomic variance.
- Results survive excluding the Great Recession and are visible in rolling-window regressions.
- VAR impulse responses show persistent declines in macro activity after oil variance shocks.
- Oil variance predicts higher oil inventories and lower oil consumption and production, supporting a precautionary-storage mechanism.
- Oil futures basis evidence connects oil variance to market prices and storage theory.
- Cross-sectional return evidence shows that non-oil industries and the aggregate market are negatively exposed to oil variance, while the oil sector is positively exposed.
- A production economy with oil inventories reproduces the qualitative empirical patterns.
- The key propagation mechanism is that oil uncertainty raises precautionary inventories, reducing effective oil input and lowering macroeconomic activity.
- The effects are stronger when oil is difficult to substitute with capital.

7 Conclusion

7.1 Core conclusion

Gao, Hitzemann, Shaliastovich, and Xu show that oil volatility risk is a distinct and economically meaningful source of macro-financial risk. Oil variance is not simply a repackaged measure of equity-market volatility or generic macro uncertainty. It forecasts future real activity, oil-sector quantities, and asset returns in ways that match a production economy with oil inventories.

7.2 Economic mechanism

The central mechanism is precautionary oil storage. A rise in uncertainty about oil supply makes future shortages more likely. To insure against these stock-out states, agents hold more oil inventories. This

increases inventories but reduces oil available for current production. Because oil is an input into final goods production and is not perfectly substitutable with capital, the fall in effective oil input lowers output, consumption, investment, and employment.

7.3 Why the evidence is convincing

The empirical evidence is persuasive because the same sign pattern appears in several places: predictive regressions, Great Recession exclusions, rolling-window estimates, VAR impulse responses, oil-sector quantities, oil futures basis exposures, sectoral dividends, and portfolio return exposures. The model then reproduces these patterns in simulated data. This agreement across data and model strengthens the interpretation that oil uncertainty has a genuine macroeconomic transmission channel.

7.4 Asset pricing implications

The paper also clarifies why oil volatility matters for asset prices. Aggregate equity and most non-oil industries are harmed by oil uncertainty because it depresses broad production and future cash flows. Oil-sector equity behaves differently because oil scarcity and inventories can raise the value of oil-related claims. This explains why oil variance exposure is negative for the broad market but positive for the oil industry.

7.5 Broader takeaway

The broader lesson is that sector-specific uncertainty can become aggregate when the sector supplies a hard-to-substitute input. Oil volatility risk is therefore not only a commodity-market variable. It is a macro-finance state variable that links production, inventories, futures markets, and equity prices.

7.6 Limitations and future research

The paper does not provide a randomized shock to oil uncertainty, so causal interpretation depends on controls, robustness, and the model. Future work could separate geopolitical, technological, and financial components of oil uncertainty; study heterogeneous firm-level exposure to oil input risk; and examine how energy transition changes the macro relevance of oil volatility.